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Veg on the menu? Differences in menu design interventions to increase vegetarian food choice between meat-reducers and non-reducers

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ABSTRACT

Reduced meat intake by Western consumers would benefit public health and the environment. However, meat consumption is notoriously difficult to change, among other things because it is often the outcome of automatic and habitual behavior. Interventions that make use of automatic decision-making processes are therefore promising, but these too vary in their success. The effectiveness of interventions may be improved if different types of meat-eaters are separated. Thus, the aim of this study was to test the effectiveness of two kinds of intervention (namely, changing to a vegetarian default on a menu and changing the attractiveness of a food label) on vegetarian food choice for two distinct groups: meat-eaters who are not reducing their meat intake (non-reducers) and meat-reducers. We also explored whether the effect of these interventions could be strengthened by activating different roles (i.e. a consumer role versus a citizen role). The online study included two European countries, Denmark ($n = 740$) and the Netherlands ($n = 749$), and involved two experiments. It was found that a vegetarian default significantly increased vegetarian food choice among the non-reducers but did not significantly do so among meat-reducers. Attractive labeling marginally increased vegetarian food choice for non-reducers but had no impact on meat-reducers. The activation of roles did not influence food choice. We conclude that meat-reducing interventions, especially where the menu default is concerned, could benefit from increased focus on a more specific target audience (i.e. non-reducers). More generally, we recommend that future meat-reducing interventions should consider different groups of meat-eaters.

1. Introduction

There is increasing consensus (Poore & Nemecek, 2018; Willett et al., 2019) that the high volume of meat consumed in Western countries has a range of negative results. Processed and red meats are (probably) carcinogenic (World Health Organization, 2015), and lead to cardiovascular disease (Abete et al., 2014; Zhong et al., 2020) and type 2 diabetes (Micha et al., 2012). Moreover, reduced meat consumption and production could lower human-produced greenhouse gas emissions significantly and would lead to more efficient use of farmland (Poore & Nemecek, 2018).

Changing meat-eating habits, however, is a notoriously difficult task as eating habits are embedded in the lifestyle of consumers and their socio-cultural environment (Vermeir et al., 2020). Furthermore, most food choices are the outcome of automatic decision-making processes (Ensaff, 2021) (meaning that the choices do not arise from conscious deliberations (Kahneman, 2011)). These automatic decision processes are steered by environmental cues (Marteau et al., 2012), such as the positioning or labeling of foods. Encouragingly, this automaticity presents an opportunity to guide eating behavior in a healthier and more sustainable direction through interventions that manipulate environmental cues.

Research on interventions designed to manipulate environmental cues to reduce meat consumption is still limited (Byerly et al., 2018), and existing studies report mixed results. On the one hand, increasing the visibility of a vegetarian dish (Kurz, 2018) and providing a vegetarian default menu (Campbell-Arvai et al., 2014) have been shown to significantly increase vegetarian food choice. On the other, making a vegetarian dish the "dish of the day" (Zhou et al., 2019) and highlighting the benefits of reduced meat consumption on a menu (Campbell-Arvai and Arvai, 2015) had no impact on vegetarian food choice.

These different outcomes could be attributable to the complexity of eating behavior, or to a lack of clarity about the mechanisms underlying environmental food choice interventions (Ensaff, 2021). Additionally,

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the meat-reducing interventions just referred to were designed according to a one-size-fits-all approach, without differentiating between different types of consumer, despite previous literature underscoring the relevance of segmentation in sustainable food consumption (Onwezen, 2018; Onwezen et al., 2012; Verain et al., 2012) and specifically in meat consumption (Dagevos, 2021; Hielkema & Lund, 2021; Lacroix & Gifford, 2020). Thus, with meat reduction, one useful research avenue would be to test the effectiveness of interventions on both consumers who have already reduced their meat intake ("meat-reducers") and on those who have not (yet) actively done so (hereafter, "non-reducers").

The effectiveness of interventions may depend not just on *who* receives them, but also on the *context* of the interventions (Byerly et al., 2018). Simply put, consumers may react differently to the same intervention in different situations. We examined two simple contextual differences: food choices made when the consumer role or citizen role is activated.

1.1. The contributions of this study

The overall aim of this online study was to determine whether the effectiveness of two meat-reducing behavioral interventions (i.e. the default and the labeling intervention) differs between meat-reducers and non-reducers, and to investigate whether the effectiveness of these interventions can be strengthened by activating the citizen versus the consumer role.

Our research contributes to the literature by (1) testing meat-reducing interventions on the two types of meat-eater we refer to as meat-reducers and non-reducers; (2) exploring the effect of two different roles (citizen versus consumer role) on food choice; (3) including large and representative samples from two European countries (Denmark and The Netherlands, countries in which the frequency of meat consumption is high (European Commission, 2013)), thereby enhancing the robustness of the results; and (4) improving the external validity of the study by including real-world consequences of online actions. We do the latter by giving participants the option to win a voucher for the foods assessed in the experiments.

1.2. Theoretical framework

Most food choices seem to be the outcome, not of careful deliberation, but of more or less automatic decision-making processes (Pohlmann, 2020). This accords with dual process theory, which describes two decision processes: one automatic, emotional, and rapid ("system 1"), and also the most frequently used, and the other reflective, rational, and slow ("system 2") (Evans, 2008; Kahneman, 2011). The automatic decision process relies on cues (Evans, 2008) which can be manipulated to generate different behavioral outcomes (Verplanken & Wood, 2006). In our study, this manipulation is called "behavioral intervention". It is also known as "nudging.".

Marteau and colleagues (2012) describe two broad categories of behavioral intervention: interventions that *alter the choice environment* and interventions that target *automatic associative processes* (Marteau et al., 2012). Placing the desired food in close proximity to the consumer (Bucher et al., 2016) is a good illustration of altering the choice environment. Automatic associative processes can be altered by, for example, describing vegetables in more indulgent terms (Turnwald et al., 2017). Our study assessed both types of behavioral intervention.

Altered defaults are in general powerful behavioral interventions. Consumers tend to accept the default option as a result of being unaware that other options are available, seeking to avoid loss aversion, experiencing inertia (unwillingness or inability to choose), or perceiving the default option to be the recommended option (Dhingra et al., 2012; Sunstein, 2017). The efficacy of defaults in steering eating behavior has been demonstrated. For example, it has been shown to increase the frequency with which wholegrain bread is chosen over white bread (van Kleef et al., 2018) and with which low-calorie dishes are chosen over high-calorie alternatives (Downs et al., 2009). Defaults also seem to reduce meat consumption. In one university-based study, a menu that offered vegetarian meals as the default increased vegetarian food choice in comparison with the non-default menu (Campbell-Arvai et al., 2014). The findings of a study of a vegetarian lunch default at conferences were similar (Hansen et al., 2021). To date, however, the effectiveness of default interventions in reducing meat consumption has yet to be tested against larger, representative samples.

Accordingly, we hypothesized the following:

H (1a): A menu with a vegetarian default option instead of a meat default option will result in a higher percentage of respondents choosing the vegetarian option.

Interventions that manipulate food labeling may alter automatic associative processes, which can in turn influence food choice. For instance, emphasizing the healthiness of a particular food lowers its taste expectations (Raghunathan et al., 2006). Explicit vegetarian labeling (i.e. highlighting that a dish does not contain meat with language such as "vegetarian" or "meat-free") can yield similar negative associations, whereas indulgent labeling (e.g. "cheesy") may have more positive associations. This tendency was observed in a field study in the UK in which indulgent labeling of vegetarian meals using phrases such as "field-grown" or "Cumberland-spiced" increased their sales in comparison with meals labeled as "meatless" or "vegetarian" (Bacon et al., 2018). However, not all indulgent labels performed equally well in the study. Additionally, although (Campbell-Arvai et al., 2014) showed that appealingly labeled vegetarian dishes, such as "three-cheese lasagna", were more frequently chosen than vegetarian dishes with unappealing labels (e.g. "vegan calzone"), the effect was not as strong as the default effect tested in the same study. More research on labeling interventions with more representative samples is needed to bring some order to these mixed results.

Thus, we hypothesized the following:

H (1b): An indulgent label, as compared with an explicit vegetarian label, will result in a higher percentage of respondents choosing the vegetarian option.

Most behavioral interventions have applied a one-size-fits-all approach, despite recommendations that interventions should be tailored in accordance with different stages of change for improved effects (Lacroix & Gifford, 2020; Lichtenstein & Glasgow, 1992). Where meat reduction is concerned, the effectiveness of interventions could be enhanced by distinguishing between meat-reducers and non-reducers.

The following theoretical considerations explain why the effectiveness of interventions might vary across these two groups. First, the groups may differ in *motivation* and *capability*, which, combined with *opportunity,* are the three relevant drivers of meat consumption behavior classified in the recent COM-B model (Graça et al., 2019; Michie et al., 2011). *Motivation* in this model encompasses both reflective (i.e. analytical decision-making) and automatic (i.e. habitual and emotional responses) forms of motivation (Michie et al., 2011). Meat-reducers presumably score higher on motivation to eat vegetarian meals than non-reducers, as they are likely to have more internal motivation to reduce their meat intake. Further, their established behavior of eating vegetarian meals probably strengthens both their reflective and automatic motivation, thereby reinforcing their intentions and habits (Bacon & Krpan, 2018; Verplanken & Orbell, 2003). Linked to this, as meat-reducers are already consciously engaging in meat-reducing behavior, they may be less affected by subconscious interventions than are non-reducers who have not yet (consciously) engaged in the desired behavior. *Capability* refers to the physical and psychological capacity to fully engage in a behavior (Michie et al., 2011). Given their already established meat reduction habits, meat-reducers are likely to be more capable physically (e.g. in terms of cooking vegetarian dishes) and

psychologically (e.g. in feeling fine about eating a food explicitly labeled as vegetarian) than non-reducers. Therefore, *opportunity,* which comprises factors external to the individual, such as environmental and social cues (Michie et al., 2011), may have a relatively stronger impact on non-reducers as their behavior is less directed by capability and motivation.

Another theoretical consideration is based upon a suggestion made by Bacon and Krpan (2018). They found that infrequent eaters of vegetarian meals were indeed more likely to choose a vegetarian option when it was indulgently labeled or marked as the chef's recommendation, but that these menu changes worked less well among frequent eaters of vegetarian meals. They argued that the second group might have exhibited "moral licensing" thinking. Moral licensing involves the belief that one can behave less morally when one has acted morally previously (Blanken et al., 2015). Eating vegetarian would probably be regarded as more moral than eating meat. Emphasizing vegetarian meals on the menu may make meat-reducers feel they have already participated in the moral behavior and thus have a "license" to behave less morally by choosing meat.

We therefore hypothesized:

H (2): Increasing the attractiveness of vegetarian options by means of (a) a vegetarian default and (b) an indulgent label is more effective in increasing vegetarian choice for non-reducers than it is for meat-reducers.

Another challenge to reducing meat consumption is presented by the fact that the attraction of eating meat – i.e. gustatory gratification – is direct and personal while the advantages of a reduced meat diet – i.e. potential long-term health benefits and environmental impacts – are more uncertain and impersonal. Most consumers do not have the foresight, or resolve, required to act on the mere promise of future benefits and instead opt for short-term personal benefits (van Kleef & van Trijp, 2018). However, in some situations, long-term benefits may be prioritized over short-term gains. This may happen when a certain role, one that values societal goals, is more salient for an individual. The sociology literature has described "the self" as a composition of multiple "role-identities" (Callero, 1985). Examples of role-identities include those of daughter, mother, colleague, and manager. Roles of this kind are always present but, depending on the context, one role-identity may be more salient than another. In most individuals, the consumer role and the citizen role coexist. We assume that an individual probably prioritizes personal goals in the consumer role and societal goals in the citizen role. Therefore, when the citizen role is activated, and is more salient to an individual, a moral imperative to behave in ways more conducive to long-term societal benefits is activated as well. By contrast, when the consumer role is activated, personal hedonic benefits become more salient, leading, in the context of our study, to a greater likelihood of choosing the meat option.

It was therefore of interest to test whether the behavioral interventions described above could be strengthened by activating the citizen role over the consumer role. In this assessment, the assumption was that the interventions might be supported by, for example, providing a description on the menu of how society could benefit from a vegetarian food choice.

Accordingly, we hypothesized:

H (3): Increasing the attractiveness of vegetarian options by means of (a) a vegetarian default and (b) an indulgent label is more effective when the citizen role, rather than the consumer role, is activated.

2. Methods

2.1. Participants

Quota sampling was undertaken by an international market research company to select the Dutch and Danish respondents. The objective here was to achieve a total sample size of 1,500 respondents, equally divided between the two countries, reflecting the age and gender distribution in the countries' populations. Sample size calculations were based on the expected incidences of vegetarian food choice found in similar studies on meat defaults and/or labeling (Bacon et al., 2018; Campbell-Arvai et al., 2014) with a p-value of 0.05 and 80% power. As the effectiveness of the default intervention was seemingly higher than that of the labeling intervention, we doubled the number of respondents in the labeling conditions. The respondents were randomly assigned to the conditions in study I (the default and labeling experiment) and the conditions in study II (activating roles in the default and labeling experiment). Respondents who ate a vegetarian, vegan, or pescetarian diet were excluded, as we focused on consumers who were able to reduce their meat consumption. The questionnaire was translated from English to Danish and Dutch by the market research company, after which it was checked by the researchers.

A total of 1,722 respondents completed the questionnaire. Respondents who were straight-liners or who answered questions unreasonably quickly were removed from the sample (78 Dutch respondents, 9.3%; 121 Danish respondents, 13.8%). Respondents who ate meat less than once per week ($n = 34$) were also excluded, as we classified them as essentially following a vegetarian diet (in line with similar studies, see Hoek et al. (2011)). The total sample comprised 1,489 respondents, of whom 749 were Dutch and 740 Danish. Socio-demographic characteristics of the respondents are shown in Table 1. Relative to the Danish and Dutch national populations, our sample was slightly older, consisted of slightly more men than women, more often lived alone, and tended to be better educated.

Table 1 also shows differences between the Dutch and Danish samples regarding socio-demographic variables, the shares of meat-reducers, levels of meat-loving attitudes (see Section 2.3) and meat frequencies. We found no significant differences in the socio-demographic variables and meat-frequencies. Danish respondents were less frequently meat-reducers and had more meat-loving attitudes than the Dutch respondents. In the subsequent analyses, we combined both respondent groups and controlled for country.

2.2. Design

We conducted two between-subject studies, each with two experiments. In study I, the participants were assigned to either the default experiment ($n = 160$) or the labeling experiment ($n = 335$). In study II, participants were assigned *either* to the default experiment, activated with either the consumer role ($n = 162$) or the citizen role ($n = 175$), *or* to the labeling experiment, activated with either the consumer role ($n = 335$) or the citizen role ($n = 322$).

In all experiments, the participants were first shown a virtual menu. After this, they were asked to make a choice between a meat burger, a vegetarian burger, or neither.

In the *default experiment*, the menu was designed in two ways: with a meat default, "Classic Burger – Beef burger with fries and salad 110DKK/€14.50" with a separate box stating "Would you rather have a Black Bean Burger? – Ask the waiter" and with a vegetarian default, "Vegetarian Black Bean Burger – Black bean burger with fries and salad 110DKK/€14.50" with a separate box stating "Would you rather have a Classic Burger? – Ask the waiter".

In the *labeling experiment*, participants were shown both burgers on the menu, but the vegetarian burger was either explicitly labeled "Vegetarian Black Bean Burger" or indulgently labeled "Flame-grilled Black Bean Burger." Our decision to use the flame-grilled label was based on the findings of a pre-study (using a convenience sample, $n = 270$) in which the popularity of several names was tested. We checked for an order effect (i.e. whether it mattered which burger was mentioned first), but no such effect was found.

In study II, we activated the participants with either the *consumer role* or the *citizen role* (see section 2.3), after which they were randomly

Table 1
Comparison of Dutch and Danish respondents in terms of socio-demographic variables, share of meat-reducers, score on meat-lover scale and frequency of meat consumption. The samples are also compared with the relevant national populations (data from DST.dk and CBS.nl).

	Danish respondents ($n = 740$)	Danish population	Dutch respondents ($n = 749$)	Dutch population	Total sample ($n = 1489$)
Age (mean, SD), NS	46.3 (16.4)	42.0	47.2 (15.6)	42.2	46.7 (16.0)
Gender (%), NS					
Men	52.8%	49.7%	51.7%	49.7%	52.2%
Women	47.0%	50.3%	48.3%	50.3%	47.7%
Other	0.1%	N/A	0%	N/A	0.1%
Education (%), NS					
Lower-level education (secondary or below)	11.8%	24.5%	12.8%	28.5%	13.0%
Middle-level education (upper secondary/post-secondary, non-tertiary)	43.9%	39.7%	43.5%	36.0%	43.7%
Higher-level education (tertiary education)	43.5%	34.6%	43.1%	33.8%	43.3%
Other/unknown education level	0.8%	1.2%	0.5%	1.6%	0.7%
% Single households, NS	57.1%	39.1%	60.1%	38.5%	58.6%
% Meat-reducer (vs. non-reducer), χ^2 (1) = 53.400, p <.001	23.5%	N/A	41.4%	N/A	
Meat-lover scale (mean, SD), t (1482.045) = -7.147, p =.000	4.8 (1.3)	N/A	4.3 (1.2)	N/A	4.6 (1.3)
Total meat consumption (red meat and poultry) per week (median, IQR), NS	4 (3–5)	N/A	4 (3–5)	N/A	4 (3–5)

assigned to either the default experiment or the labeling experiment (as described above).

2.3. Measurements

Demographics. All participants were asked to verify their age, gender, educational level, and household composition at the start of the survey. Educational level was recoded into three levels, and other ($n = 10$) was combined with the lowest level of education due to the small sample size and to avoid missing values.[1]

Behavioral choice task. The participants were shown one of the menus (see Section 2.2) and then asked to choose either one of the two presented burgers or neither. Afterwards, they were asked to indicate, on a 7-point Likert scale ranging from 1 (very unlikely) to 7 (very likely), how likely they would be to make the same choice in a real restaurant and whether they would be likely to do so in the next three weeks. The results showed that most participants would probably make the same choice in a real restaurant: classic burger (*M (SD)* = 5.0 (1.8)); vegetarian burger (*M* (*SD*) = 5.4 (1.3)). They were, however, slightly less likely to do so within the next three weeks: classic burger (*M (SD)* = 4.4 (1.9)); vegetarian burger (*M (SD)* = 5.0 (1.4)).

Ingredients check. We checked to determine whether any of the participants had an intolerance, allergy, religious objection, or other aversion to the burgers' ingredients that would prohibit them from choosing freely among them. Participants with an aversion stated that they did not want to eat either of the burgers.

Meat-reducers and non-reducers were assessed by asking respondents whether they had plans to reduce their meat intake or had already reduced their meat intake, in line with (Hielkema and Lund, 2021). The answer options corresponded to the stages of change model (Prochaska et al., 1993; Prochaska & DiClemente, 1983): no intention to reduce meat intake, recent increase in meat intake, intention to reduce in the distant future (within six months), intention to reduce in the near future (<1 month), recent reduction in meat intake (within the last six months), and reduction over a longer period (more than six months). Respondents confirming one of the last two options were classified as meat-reducers; those confirming the previous stages were classified as non-reducers.

Meat-loving was included to examine whether non-reducers and meat-reducers and the Danish and Dutch respondents differed in their attitudes toward meat. It was assessed by two constructs derived from Lacroix and Gifford (2019) "liking the taste of meat" and "dependence on meat." Further, "liking the taste of meat" was measured by "meat is delicious", "meat adds so much flavor to a meal it does not make sense to leave it out", and "vegetarian food is bland and boring". "Dependence on meat" was measured by "I would feel fine with a meatless diet" "if I was forced to stop eating meat, I would feel sad" and "meat is irreplaceable in my diet". The reliability of these scales was, respectively, questionable (Cronbach's $\alpha = 0.676$) and acceptable (Cronbach's $\alpha = 0.785$). A multicollinearity check showed that the scales were highly correlated ($r = 0.727$, $p <.000$), and a principal component analysis indicated that all six items loaded on one factor. We therefore merged the two constructs into one overarching construct: "meat-loving." This scale had good reliability (Cronbach's $\alpha = 0.846$), which did not improve if items were removed.

Meat consumption frequency was assessed by asking respondents how many days per week they typically ate red meat and poultry on a scale from "never" to "seven days per week." We then recoded this measurement into an ordinal variable to match the average frequency per week (e.g. never = 0, and 3–4 times per week = 3.5) and calculated a total meat consumption variable. Fish was not classified as meat, as it can also be seen as a meat replacement product (de Boer et al., 2020).

Real-life voucher choice. To attach real-life consequences to online food choices, we designed a lottery that offered respondents the chance of winning a voucher for either a black bean burger or a classic burger. They were also given the option of not participating in the lottery. For practical reasons, at the end of the survey we informed the respondents that the awarded voucher was generic and could be used to buy several types of foods.

Activation of roles (only in study II). Respondents in the second study were randomly assigned to activated consumer or activated citizen roles. Directly after answering the socio-demographic questions, the respondents rated, on four 7-point Likert scales, how important it was to eat food that was: healthy, environmentally-friendly, organically produced, and contained less meat. In the citizen role, the respondents were presented with the statement: "As a citizen, I find it important/not important for society as a whole to eat foods that …" In the consumer role, they were presented with: "As a consumer, I find it important/not important to me personally to eat foods that …" Adding subtle situational cues to the phrasing of questions has been considered adequate for making an "individualistic, relational, or collectivistic mindset accessible with consequences for identity content, cognitive procedures, and action" (Oyserman et al., 2014). Unfortunately, when we prompted the respondents with a follow-up question about whether they had answered as a consumer or as a citizen, only 41.9% stated correctly that they had answered as a citizen. The share answering correctly that they had answered as a consumer was much higher (80.3%).

Neutralizing activation. To inactivate the different roles, which could

[1] The results obtained were the same when "other" was treated as a missing value.

have inadvertently interfered with the second survey component, we asked the respondents to rate the attractiveness of eight random pictures.

2.4. Data analyses

All analyses were conducted using SPSS version 28 for Windows. We compared the Dutch and Danish samples using *t*-tests (for continuous variables), chi-square tests (for nominal variables), and Mann-Whitney U tests (for ordinal variables). The same tests were used when comparing non-reducers with meat-reducers.

We adopted the following analytical strategy to test the first and second hypotheses (study I). For both the default and labeling experiments, we analyzed the effectiveness of the conditions on the whole sample, and then on non-reducers and meat-reducers separately, in two steps. In step 1, we ran multinomial logistic regressions to test the sole effect of the conditions (independent variable (IV): vegetarian versus meat default or indulgent versus explicit label) on food choice (dependent variable (DV): vegetarian burger or neither option versus meat burger). In step 2, we conducted a robustness test by adding the control variables country, age (in years), gender (male/female), and education (three levels) as additional IVs.

For the third hypothesis (study II), we tested the effect of roles within each condition separately (meat default, vegetarian default, explicit label, and indulgent label). We ran multinomial logistic regressions with food choice as a DV and activation (consumer or citizen role) as an IV *first* for the whole sample and *then* for non-reducers and meat-reducers separately. We also included an extra robustness step with the same control variables as in study I.

For all regressions, we reported the Wald χ^2 and odds ratios (ORs), including the corresponding confidence level (CI).

Chi-square analysis was used to test whether the respondents chose the same type of food for the voucher as they did from the menu.

3. Results

3.1. Randomization check

We checked whether the respondents were equally divided over all conditions using chi-square analyses (gender, education, country, non-reducers/meat-reducers) and ANOVA (age). We found no significant differences between the conditions, except for gender in study II. A higher share of male respondents (62.3%) was observed in the indulgent labeling condition: χ^2 (5) = 12.578, p <.05.

3.2. Comparing the characteristics of non-reducers and meat-reducers

Table 2 shows that in both experiments the shares of respondents who were female, had a university education, and were Dutch, were larger among meat-reducers than they were among non-reducers. Non-reducers ate meat more frequently and scored higher than meat-reducers on the meat-lover scale. Meat-reducers in the labeling experiment were slightly older.

Table 2
Differences in characteristics of non-reducers and meat-reducers for the default experiment and the labeling experiment.

Default experiment (n = 497)			
	Non-reducers (*n* = 328)	**Meat-reducers (*n* = 169)**	**Difference tests**
Age, mean (SD)	46.0 (15.8)	46.5 (16.2)	*NS*
Gender, % female	43.0%	55.6%	*χ2 (1) = 7.142, p <.05*
Education, % with a university degree	40.6%	50.0%	*χ2 (1) = 3.962, p <.05*
Household composition, % living in single households	57.0%	62.1%	NS
Country, % Dutch respondents	44.5%	62.7%	*χ2 (1) = 14.796, p <.001*
Days per week meat consumed, median (IQR)	5 (3–7)	3 (2–5)	*U = 19804.00, p <.001*
Meat-lover scale, 1 (low score) to 7 (high score), mean (SD)	5.0 (1.2)	3.7 (1.1)	*t* (495) = 11.977, p <.001
Labeling experiment (n = 992)			
	Non-reducers (*n* = 677)	**Meat-reducers (*n* = 315)**	**Difference tests**
Age, mean (SD)	46.1 (15.9)	48.8 (16.1)	*t (990) = -2.496, p <.05*
Gender, % female	42.5%	59.4%	*χ2 (1) = 24.668, p <.001*
Education, % with a university degree	39.3%	52.6%	*χ2 (1) = 15.226, p <.001*
Household composition, % living in single households	60.0%	55.6%	NS
Country, % Dutch respondents	43.3%	64.8%	*χ2 (1) = 39.685, p <.001*
Days per week meat consumed, median (IQR)	5 (3–7)	3 (2–5)	*U = 73611.00, p <.001*
Meat-lover scale, 1 (low score) to 7 (high score), mean (SD)	5.0 (1.1)	3.7 (1.8)	*t* (990) = 16.441, p <.001

Note: NS = Not significant. SD = standard deviation.

3.3. Study I, part A: Default experiment

3.3.1. Combined sample in default experiment

First, we tested the sole effect of the default conditions on food choice for the combined sample (n = 160). The model was significant: χ^2 (2) = 8.104, p <.05, Cox and Snell R^2 = 0.049, Nagelkerke R^2 = 0.061. A vegetarian default (reference: meat default) made it 3.3 times more likely (OR (CI) = 3.27 (1.31–8.14), Wald χ^2 (1) = 6.47, p <.05) that a vegetarian burger would be chosen rather than a meat burger.

Then, we ran a robustness check by including the control variables. The model was still significant: χ^2 (12) = 54.683, p <.001, Cox and Snell R^2 = 0.289, Nagelkerke R^2 = 0.358. In consequence, it was 3.6 times more likely (OR (CI) = 3.64 (1.41–9.42), Wald χ^2 (1) = 7.12, p <.01) for a vegetarian burger to be chosen in the vegetarian default condition. Regarding the control variables, a Dutch respondent was 2.4 times more likely (OR (CI) = 2.44 (1.04–5.69), Wald χ^2 (1) = 4.22, p <.05) to choose a vegetarian burger. Further, higher age was associated with a higher likelihood (OR (CI) = 1.11 (1.04–1.18), Wald χ^2 (1) = 9.79, p <.01) of choosing neither option (reference: meat burger). Male respondents were less likely (OR (CI) = 0.12 (0.03–0.53), Wald χ^2 (1) = 7.97, p <.01) to choose neither option.

3.3.2. Non-reducers and meat-reducers in default experiment

To address the second hypothesis we ran multinomial regressions again while splitting the data between non-reducers (n = 108) and meat-reducers (n = 52).

For non-reducers, the model was significant: χ^2 (2) = 9.863, p <.01, Cox and Snell R^2 = 0.087, Nagelkerke R^2 = 0.144. For meat-reducers, the model was not significant: χ^2 (2) = 2.040, NS, Cox and Snell R^2 = 0.038, Nagelkerke R^2 = 0.044. The model remained significant for non-reducers after the robustness check: χ^2 (12) = 31.660, p <.01, Cox & Snell R^2 = 0.254, Nagelkerke R^2 = 0.421. The model was also significant for meat-reducers: χ^2 (12) = 28.202, p <.01, Cox & Snell R^2 = 0.419, Nagelkerke R^2 = 0.475. However, this was unrelated to the default conditions. Higher age made it more likely (OR (CI) = 1.09 (1.00–1.18),

Wald χ^2 (1) = 4.05, p <.05) that neither option was chosen[2].

Fig. 1 shows that a higher share of respondents chose the vegetarian burger compared to the meat burger in the vegetarian default condition. For non-reducers this likelihood without the robustness check was marginally significant (OR (CI) = 7.59 (0.92–62.31), Wald χ^2 (1) = 3.60, p =.059) and with the robustness check significant (OR (CI) = 10.09 (1.10–93.37), Wald χ^2 (1) = 4.18, p <.05). For meat-reducers, this higher likelihood was not significant, respectively without and with robustness check: OR (CI) = 2.63 (0.67–10.35), Wald χ^2 (1) = 1.90, p =.168 and OR (CI) = 3.59 (0.77–16.73), Wald χ^2 (1) = 2.66, p =.103.

It also shows that the vegetarian default led to the relatively largest increase in vegetarian food choice among non-reducers (from 2.3% to 14.1%; relative change of 513.0% and absolute change of 11.8%) and the largest absolute increase among meat-reducers (from 33.3% to 52.9%; relative change of 58.9% and absolute change of 19.6%).

3.4. Study I, part B: Labeling experiment

3.4.1. Combined sample in the labeling experiment

Both the sole effect of the labeling conditions and the robustness check led to insignificant models for the combined sample: χ^2 (2) = 3.278, NS, Cox & Snell R^2 = 0.010, Nagelkerke R^2 = 0.014 and χ^2 (12) = 12.325, NS, Cox & Snell R^2 = 0.036, Nagelkerke R^2 = 0.050.

3.4.2. Non-reducers and meat-reducers in the labeling experiment

The sole effect of the conditions led to a marginally significant model for non-reducers, χ^2 (2) = 5.516, p =.063, Cox & Snell R^2 = 0.025, Nagelkerke R^2 = 0.040, and no significant model for meat-reducers, χ^2 (2) = 0.339, NS, Cox & Snell R^2 = 0.003, Nagelkerke R^2 = 0.002. Both robustness checks led to insignificant models: non-reducers: χ^2 (12) = 17.777, NS, Cox & Snell R^2 = 0.078, Nagelkerke R^2 = 0.124; meat-reducers: χ^2 (12) = 8.378, NS, Cox & Snell R^2 = 0.070, Nagelkerke R^2 = 0.087.

Fig. 2 shows that, in the combined sample and among non-reducers, the black bean burger was chosen more frequently when it was labeled indulgently.

3.5. Study II: Activating citizen and consumer roles

We tested the effect of the two activated roles under all four conditions using multinomial logistic regressions *first* for the whole sample and *then* for non-reducers and meat-reducers. As the results were the same before and after the robustness check, only the results from the robustness check are reported.

3.5.1. Activating roles in the default experiment

Both for the whole sample and, separately, for non-reducers and meat-reducers, we found no influence of the activated roles on the effectiveness of the default conditions on food choice. The control variables age and education did influence food choice. For non-reducers within the vegetarian default,[3] we found that an increase in age led to a lower likelihood (OR (CI) = 0.94 (0.90–0.98), Wald χ^2 (1) = 8.90, p <.01) of choosing the vegetarian burger. Also, for non-reducers within the meat default,[4] higher age led to a lower likelihood (OR (CI) = 0.86 (0.74–0.98), Wald χ^2 (1) = 4.88, p <.05) of choosing the vegetarian burger. An upper secondary/post-secondary education level (reference: tertiary education) also made it less likely (OR (CI) = 0.05 (0.00–0.92), Wald χ^2 (1) = 4.06, p <.05) that non-reducers within the meat default would chose a vegetarian burger.

3.5.2. Activating roles in the labeling experiment

Other than for non-reducers in the indulgent labeling condition, we found no effect of the activated roles in the labeling conditions.[5] Non-reducers in the indulgently labeled condition[6] with an activated citizen role (reference: consumer role) were 3.6 times more likely (OR (CI) = 3.64 (1.31–0.10), Wald χ^2 (1) = 6.15, p <.05) to choose the vegetarian burger (reference: meat burger). Fig. 3 illustrates this. Higher age also made choosing a vegetarian burger less likely (OR (CI) = 0.96 (0.94–0.99), Wald χ^2 (1) = 5.71, p <.05). An upper secondary/post-secondary education level (reference: tertiary education) made choosing a vegetarian burger 3.5 times more likely (OR (CI) = 3.49 (1.07–11.40), Wald χ^2 (1) = 4.28, p <.05). We also found that, within the explicit labeling condition[7] and in the whole sample, higher age made choosing a vegetarian burger less likely (OR (CI) = 0.97 (0.94–0.99), Wald χ^2 (1) = 8.43, p <.01).

3.6. Voucher choice

Finally, we checked whether the fictive choice of a vegetarian or classic burger was associated with real-life voucher choices. We found that the type of menu choice made choosing the same voucher type significantly more likely: total sample, χ^2 (4) = 523.294, p <.001; non-reducers, χ^2 (4) = 270.621, p <.001; meat-reducers, χ^2 (4) = 173.581, p <.01. Fig. 4 shows that, of the total sample, the majority (67%) who chose the vegetarian burger also chose the vegetarian voucher. This percentage was higher for meat-reducers (75%) and lower for non-reducers (56%).

4. Discussion

Our results showed that a vegetarian default increases vegetarian food choice, in line with previous findings (Campbell-Arvai et al., 2014; Hansen et al., 2021). We found that the vegetarian default led to the greatest relative increase in vegetarian food choice among non-reducers (a relative increase of 513% compared to 59% among meat-reducers). Thus we were able to confirm our first and second hypotheses in relation to the default experiment. We could not confirm them, however, in relation to the labeling intervention, although we did find a marginal effect that an indulgent label led to an increase in vegetarian food choice among non-reducers. Nor could we fully confirm our third hypothesis, as we only found an effect of the activated roles for non-reducers in the indulgent labeling condition.

Our default intervention had a large effect size for non-reducers and a medium effect size for all meat-eaters (see Chen and colleagues (2010) for effect size interpretation of ORs). The default intervention, in line with our expectations, had no significant effects on meat-reducers. Non-reducers may have been affected more by the interventions, as meat-

[2] To assure the differences in the significance of the regression models between non-reducers and meat-reducers were not solely based on a difference in sample size, we ran an additional regression analysis on a random sample of 52 non-reducers. The model was marginally significant χ^2 (2) = 5.900, p =.052, Cox & Snell = 0.107, Nagelkerke = 0.192. Due to none of the 52 non-reducers choosing the vegetarian burger in the meat default, odds ratios could not be meaningfully calculated.

[3] Model fit: χ^2 (12) = 24.548, p <.05, Cox and Snell R^2 = 0.106, Nagelkerke R^2 = 0.155.

[4] Model fit: χ^2 (12) = 26.357, p <.05, Cox and Snell R^2 = 0.196, Nagelkerke R^2 = 0.348.

[5] As more male respondents were present in the indulgent labeling condition (see section 3.1), we controlled for gender by including it as an interaction effect with the conditions via step-wise analyses; the interaction was not significant and therefore not included in the model.

[6] Model fit: χ^2 (12) = 30.362, p <.01, Cox and Snell R^2 = 0.120, Nagelkerke R^2 = 0.189.

[7] Model fit: χ^2 (12) = 34.528, p <.01, Cox and Snell R^2 = 0.104, Nagelkerke R^2 = 0.151.

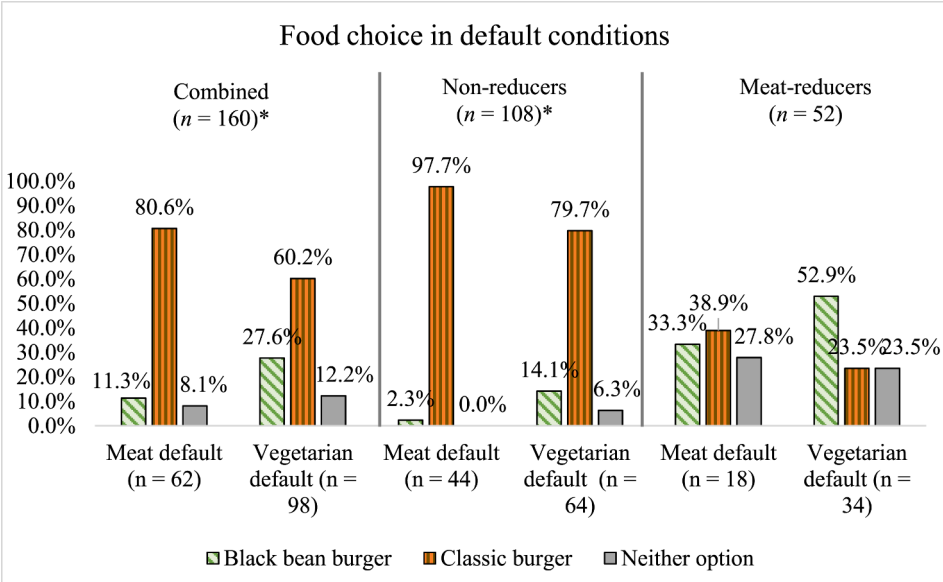

Fig. 1. Share of respondents who chose the vegetarian burger, the meat burger, or neither option in the meat or vegetarian default condition; * indicates significant differences in the default conditions concerning food choice.

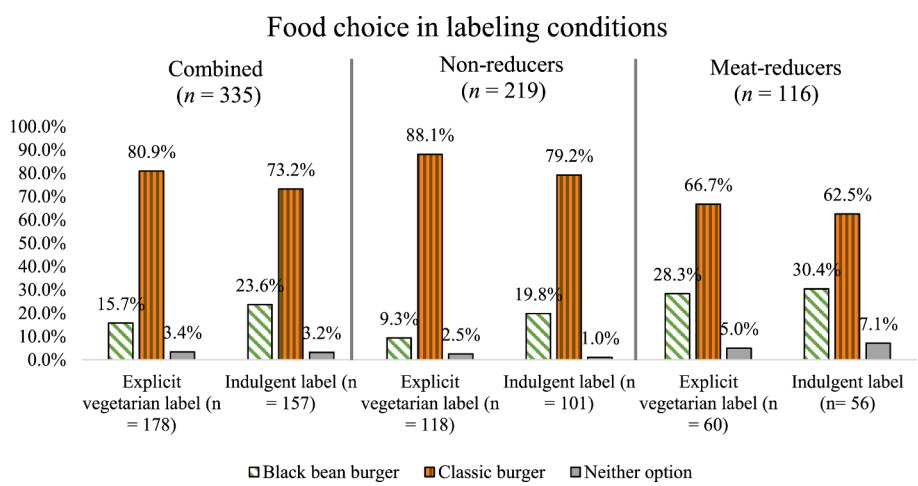

Fig. 2. Food choice in the different labeling conditions of the black bean burger, with no significant differences found.

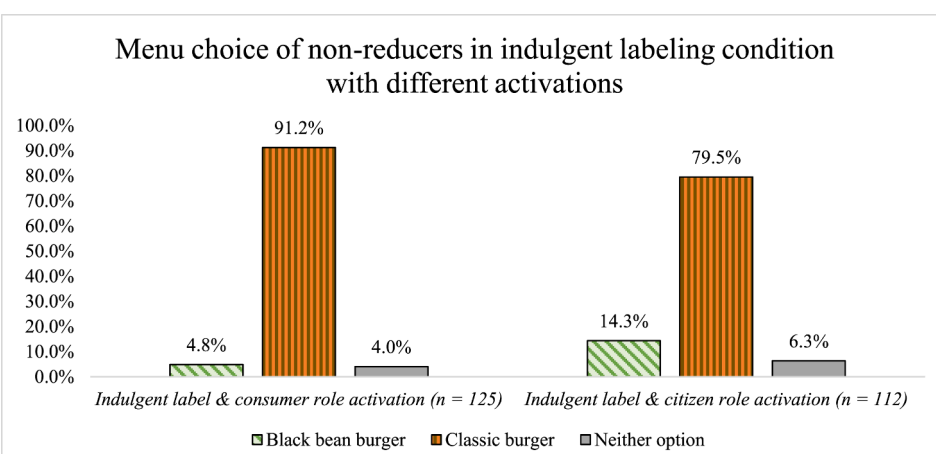

Fig. 3. Menu choice of non-reducers in the indulgent labeling condition with an activated citizen or consumer role.

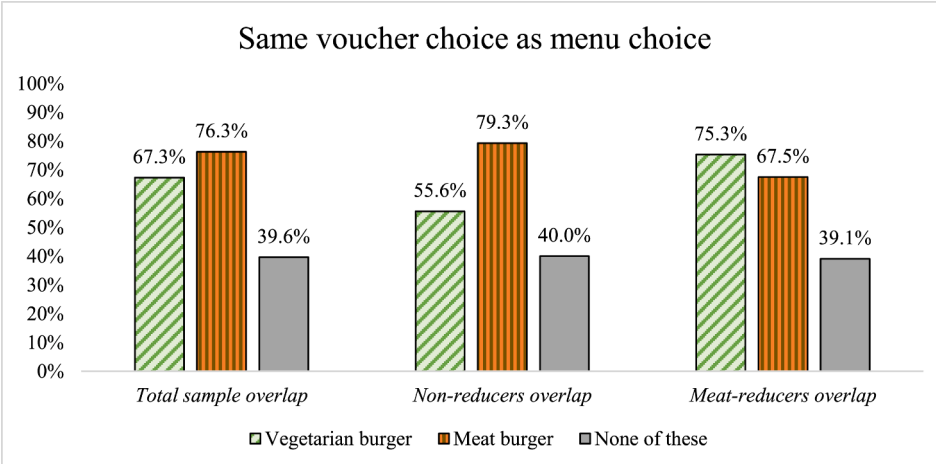

Fig. 4. Share of respondents who chose the same type of burger on both the menu and when presented with a voucher.

reducers have relatively less to "gain" as they already eat more vegetarian meals. A similar study also observed the steepest increase in vegetarian consumption among consumers with the lowest previous vegetarian food intake (Garnett et al., 2019). Another explanation is that these interventions mostly work via automaticity, which may be why the largest effects can be observed among those who do not yet consciously perform the desired behavior – in this case, non-reducers. Further, the opportunity component of behavior may have played a relatively larger role among non-reducers. Lastly, meat-reducers may have displayed "moral licensing".

We found no significant effects of indulgent labeling on vegetarian food choice. At most, we found a marginal effect for non-reducers without controlling for other variables. Potentially, then, labeling is not a sufficiently strong intervention on its own. Campbell and colleagues (Campbell-Arvai et al., 2014) found weaker effects of labeling interventions in comparison with a default intervention. Potentially, labeling interventions may become more successful when consumers become more familiar with the indulgent labels and vegetarian food.

We found that activating the citizen role *and* indulgently labeling the vegetarian option increased vegetarian food choice among non-reducers (with a medium effect size). However, it should be noted that, in study I, where there was no such activation, the choice of the vegetarian burger was even higher (19.8% compared to 14.3%). Thus, it is possible that rather than citizen activation exerting a strengthening effect, consumer activation generates negative effects (here, the vegetarian choice was only 4.8%). We must also acknowledge that the activation of neither role was fully successful: our manipulation check showed that the majority of respondents in the activated citizen role stated that they answered the activation questions in the consumer role.

Most respondents stated that they chose a voucher with the same food choice as the one they had made on the menu. This indicates that the choices observed in our experiment spilled over to offline choices, thereby yielding real consequences. This is in line with a recent experimental study that found no differences in outcome when an online choice or a voucher choice were included as dependent variables separately or in combination (Asbridge et al., 2021).

Our results were made more robust by the inclusion of two countries. In some cases, the Dutch respondents were more likely than the Danish ones to choose the vegetarian burger over the meat burger. We assume that this may be due to the fact that there was a larger share of meat-reducers among the Dutch respondents.

4.1. Recommendations

The vegetarian default resulted in 14% of non-reducers choosing a vegetarian burger, in contrast with 2% in the meat default. One way to interpret these results is by saying that 12 out of 100 burgers were additionally vegetarian instead of beef burgers. A typical 4 oz. beef patty leads to the emission of roughly 20 kg CO^2 equivalent (CO^2e), while a typical vegetarian patty only leads to 3 kg CO^2e (Saget et al., 2021). For every 100 burgers sold, a reduction of 204 kg CO^2e could result, the equivalent of an average car driving from Amsterdam to Copenhagen (EPA, 2021). If these choices reflect real behavior, then changing to a vegetarian default would have an undeniably significant impact.

Therefore, we suggest that food outlets such as restaurants and canteens should carefully consider the type of dish they treat as the default. They should opt for vegetarian defaults if they want to stimulate vegetarian consumption. Changing to a vegetarian default is a low-cost policy intervention, and one that would not interfere with the availability of food types and would as such have only a limited impact on business models (Ensaff, 2021). Furthermore, in contrast with a meat tax, defaults do not differentially affect different socioeconomic groups. A vegetarian default menu might also be a better option than forced vegetarian days/menus at institutions: in one study, an obligatory vegetarian day at a school led to non-compliance (Lombardini & Lankoski, 2013)).

Although indulgent labeling marginally increased vegetarian food choice among non-reducers, we obtained no significant results on indulgent versus explicit labeling. However, as previous studies have found a significant increase in vegetarian food choice when such foods are indulgently labeled (Vennard et al., 2019) and when explicit vegetarian labeling is avoided (Bacon & Krpan, 2018), we suggest that indulgent labeling could be used as a supportive measure, perhaps in combination with a vegetarian default (Campbell-Arvai et al., 2014). We further suggest that activating personal hedonic goals, such as a consumer role, is best avoided, as this seems to deter vegetarian food choice.

4.2. Limitations and suggestions for future research

Our study had several limitations. First, the sample size of meat-reducers in the default study was low. This may have led to type II error. In addition, we measured self-reported behavior, yet respondents may behave differently in reality. Further, as a result of the online set-up actual consumption was not measured. However, it has been argued that food choices, in contrast with actual food consumption, can be used to test the impact of interventions (Cadario & Chandon, 2020; Ensaff, 2021). Our menu was also simplified (i.e. offered two options); effects may be different with larger menus with more options, although in fact larger menus in other studies yielded similar outcomes (Campbell-Arvai et al., 2014; de Vaan et al., 2019). Another observation was that meat-reducers chose more frequently for neither option (especially in the default experiment). Although we do not know the reasons why they did

so, this might indicate that for this group of consumers both burgers were not attractive. Also, although we tested two countries, Denmark and the Netherlands are in many respects similar, and therefore generalization to other parts of the world is limited.

We recommend that future research should test the long-term effects of these interventions. Currently, there is mixed evidence on the effectiveness of behavioral interventions over the long term (Marchiori et al., 2017), and their effects on vegetarian consumption remain understudied. Further, is it likely that behavioral interventions are context-specific and do not transfer readily to other situations (e.g. in other restaurants or at home). Future research could examine whether these interventions lead to spillover effects, inspiring consumers to eat similar vegetarian foods at home (Hansen et al., 2021) or to pursue a more plant-based diet in the future (Bianchi et al., 2018). It would also be well worth investigating whether vegetarian default interventions lead to reactance (Brehm, 1966; de Vaan et al., 2019) or compensation behavior (De Boer et al., 2014). In the case of reactance, meat-eaters would feel that they are being pushed toward a vegetarian choice, and consequently that their freedom is being threatened, which may lead to them *not* choosing the vegetarian option. In the case of compensation behavior, they would eat more meat at a later point.

To improve our understanding of behavioral interventions, future research will ideally focus on the mechanisms underlying the interventions. They also need to pay attention to differences in effectiveness for non-reducers and meat-reducers.

Future research could also assess the effect of voucher choice (as this was shown to be a promising supportive tool) by including it in the experimental design (e.g. by making it the dependent variable, see Asbridge et al., (2021)).

In addition, future research could test whether our finding that individuals who do not yet consciously perform the desired behavior are relatively more impacted by behavioral interventions than consumers who have already changed their behavior is observable in other settings and conditions (e.g. healthy eating or food waste interventions).

4.3. Conclusion

We showed that vegetarian default interventions increases vegetarian consumption relatively the most among those who have not yet reduced their meat intake. We therefore recommend that behavioral interventions are designed for, and target, specific groups of meat-eaters.

CRediT authorship contribution statement

Marijke H. Hielkema: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization, Project administration. **Marleen C. Onwezen:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Machiel J. Reinders:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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